

AADI ARYA

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Education

McMaster University

Honours Bachelor of Computer Science

Present

Hamilton, Canada

- Relevant Coursework: AP Statistics, AP Computer Science, AP Calculus BC, AP Physics 2, Computer Engineering.

Experience

Associate Software Engineering Intern

Jun 2025 – Sep 2025

Canadian Imperial Bank of Commerce (CIBC)

Toronto, ON

- Contributed to an automated risk assessment tool by writing Python and SQL scripts that categorized credit applications, supporting a pilot that cut manual loan review time by an estimated **60%** for the team.
- Assisted a senior engineer in testing and refining a BERT based classifier used to flag risk indicators in financial documents, running validation checks against staging data under their guidance.
- Built a front end component of a real time monitoring dashboard in React, working alongside the team's Jenkins based deployment and unit test process.

Software Engineer

Sep 2026 – Present

McMaster RoboMaster

Hamilton, ON

- Developing real-time computer vision and autonomy software in **Python and C++** with **OpenCV** for competition robots in the RoboMaster robotics competition.
- Working in a shared Git codebase with the Autonomy/CV sub-team, integrating perception and navigation software with the embedded and mechanical sub-teams.

Developer

Sep 2026 – Present

Google Developer Groups (GDG) on Campus, McMaster University

Hamilton, ON

- Develop and maintain features for GDG McMaster's website using **HTML, CSS, and JavaScript**, turning event campaigns and branding into **responsive, accessible** pages for technical workshops and events.
- Collaborate with student developers on **open source projects**, contributing code through pull requests and code reviews on GitHub.

STEM Workshop Organizer

Jun 2024 – Jun 2026

Peel District School Board

Peel Region, ON

- Designed and delivered a hands-on software development curriculum where students built full applications using **React, Flask, TensorFlow, and AWS**.
- Mentored **100+** students in **Python, JavaScript, and C++**, and led workshops on circuit building and hardware-software systems.

Personal Projects

QuantAlpha: Algorithmic Trading Simulation Engine | *Python, scikit-learn, pytest, Docker*

- Simulated multi-exchange market microstructure (latency, noise, liquidity events) to generate realistic tick-level order flow.
- Built tick-by-tick arbitrage detection filtered by fee/slippage thresholds, executing paper trades on simulated data.
- Trained **GradientBoostingClassifier** and **Ridge** models on lagged returns, reaching **~65–67% held-out accuracy**.
- Validated with a **24-test** pytest suite and Docker packaging; documented the paper-trading model's limitations honestly.

NeuroScribe: Synthetic Medical Imaging Segmentation Pipeline | *PyTorch, FastAPI, React, AWS S3, Docker*

- Generated synthetic 2D scans with lesion-like blobs and matching ground-truth masks.
- Trained a **U-Net** (SmallUNet) with combined BCE and Dice loss, cutting training loss from **1.43 to 0.34** over **20 epochs**.
- Added gradient-based interpretability (saliency and Grad-CAM-style heatmaps) via a FastAPI **/segment** endpoint.
- Built S3-compatible storage (boto3, tested with **moto**) and a pytest suite; documented as a non-clinical prototype.

Skills

Languages: Python, C++, JavaScript, SQL, HTML/CSS, Java, C, Bash

AI/ML: PyTorch, TensorFlow, scikit-learn, OpenCV, BERT, Computer Vision

Web & Cloud: React, Next.js, FastAPI, Flask, Node.js, MySQL, AWS (S3), Docker

Tools: Git, Linux, pytest, Jenkins, Jira, Agile

Competitions/Certificates: Euclid/CCC Waterloo Math Contests (**top 15%**), DECA Internationals, Harvard CS50